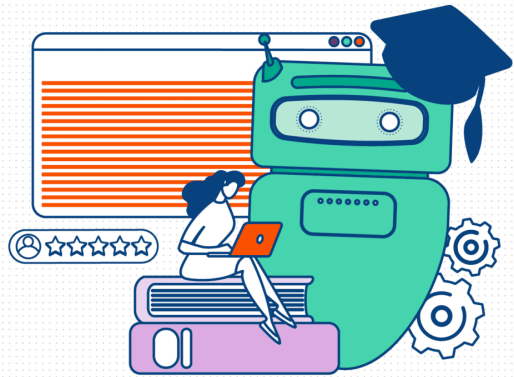


PVMark: Enabling Public Verifiability for LLM Watermarking Schemes

Haohua Duan, Liyao Xiang*, Xin Zhang, Baochun Li, Bo Li

Tracing LLM-Generated Text Is Essential for Responsible AI

Risks of LLM Misuse



Academic Misconduct



AI-Generated Fake News

Responsible AI requires reliable provenance tracing for LLM-generated text.

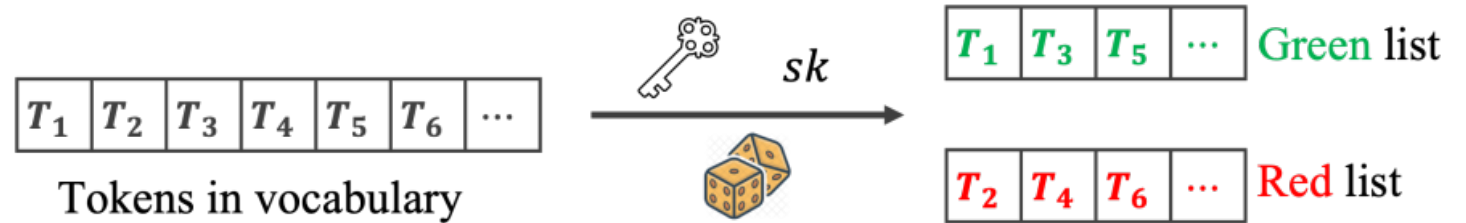
LLM watermarking embeds detectable signals into generated text, offering a practical approach to provenance tracing.

Probability-Shift LLM Watermarking

Representative Example: KGW (Kirchenbauer et al., 2023)

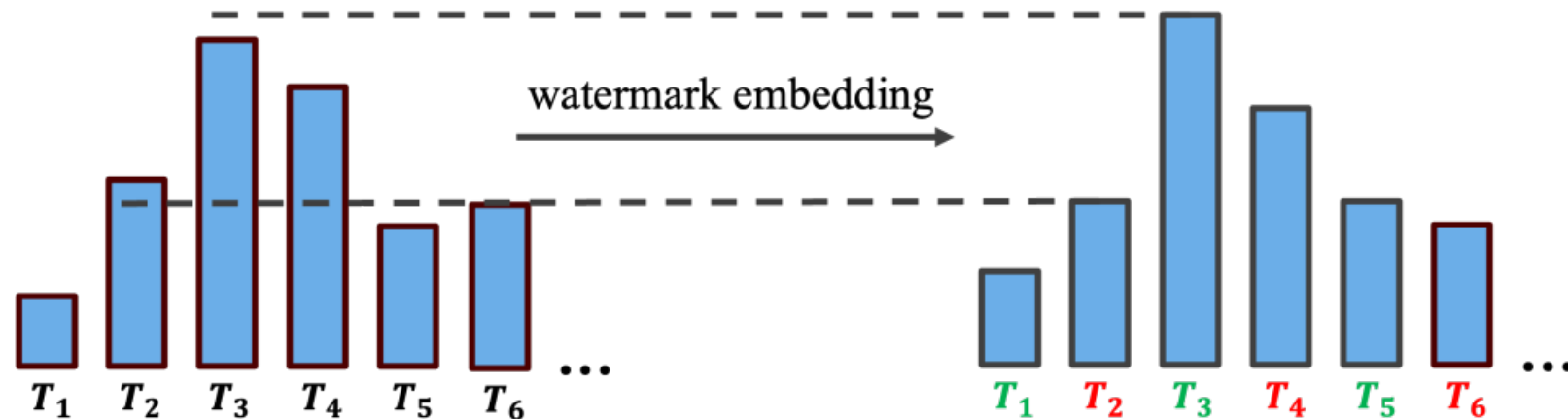
Embedding Process :

- ① Random seed generation
- ② Vocabulary Partitioning
- ③ Probability shifting and sampling



Detection Process :

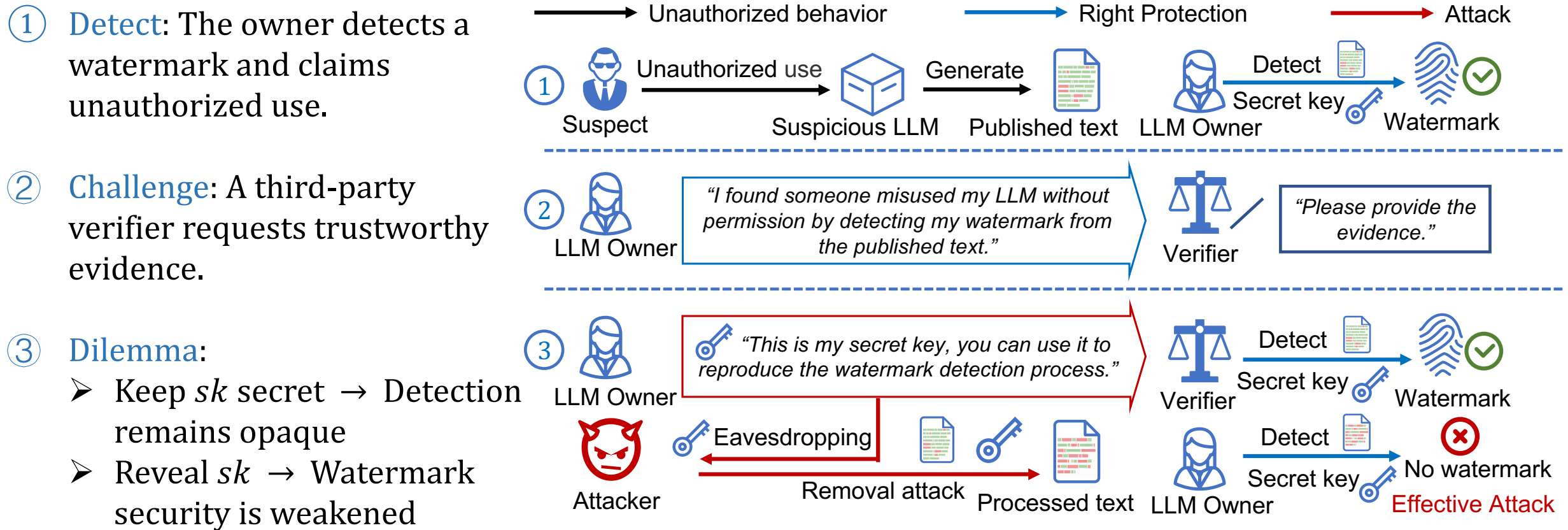
- ① Compute random seed
- ② Check and count green tokens
- ③ Compute watermark score



Embedding and detection are coupled through the same secret key.

Watermark Detection Creates a Trust Dilemma

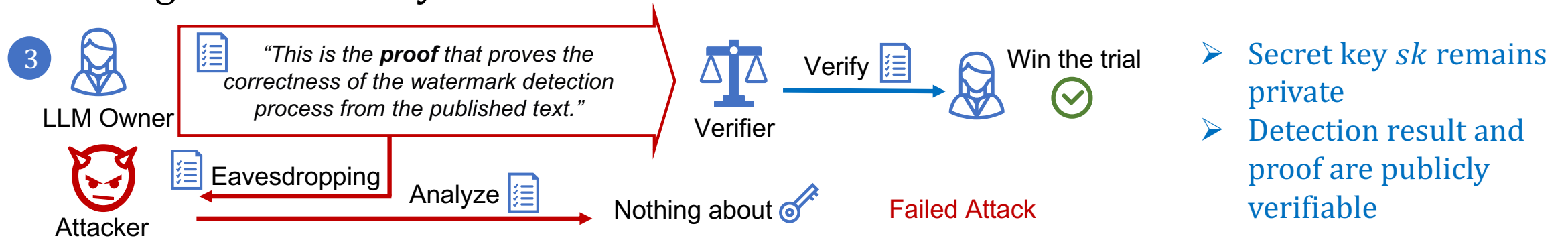
Watermark Security v.s. Public Verifiability



The key must remain secret, while detection must be publicly verifiable.

Our Work——PVMark: Private Detection, Public Verification

PVMark provides a **correctness proof of watermark detection**, rather than revealing the secret key.



Core idea: Integrate zero-knowledge proofs (ZKP) ^① into existing watermark detection.

Two Technical Challenges:

➤ Make Detection Provable:

Translate watermark detection into ZKP-friendly circuits. Reconstruct the detector using lookup, hashing, comparison, and aggregation constraints.

➤ Make Proof Generation Efficient:

Reduce proof costs by exploiting the structure of watermark detection. Reduce circuit constraints and fold repeated token-wise computations.

① ZKP allows a prover to demonstrate that a statement is true without revealing any information besides its truth.

PVMark Adds Public Verifiability while Preserving Watermark Performance

Representative results on KGW

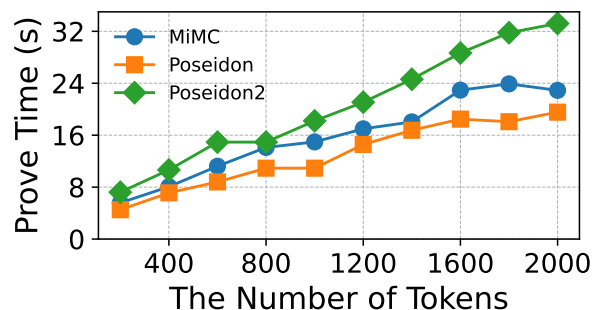
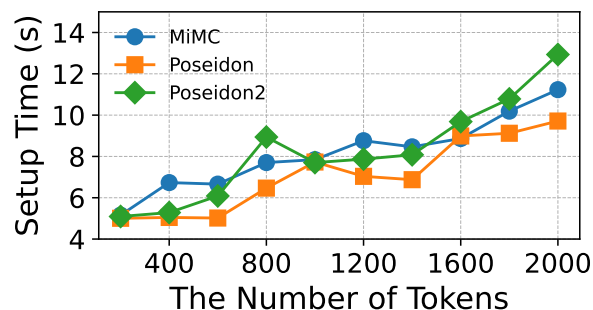
➤ Effectiveness & fidelity: Success rate (SR) ↑ / Watermark Score ↑ / PPL ↓ **PVMark**

Watermarking Scheme	Model	Dataset	Benchmark	MiMC		Poseidon		Poseidon2
				two-to-one	three-to-one	two-to-one	three-to-one	two-to-one
KGW	OPT-1.3B	C4	99% / 9.78 / 11.84	99% / 9.86 / 12.88	97% / 9.75 / 13.99	96% / 10.86 / 11.22	96% / 10.45 / 13.97	99% / 9.40 / 12.97
		Pile	87% / 10.14 / 9.46	88% / 10.03 / 10.10	94% / 11.66 / 9.87	87% / 10.32 / 9.63	94% / 11.59 / 10.41	86% / 9.79 / 23.62
	T5-Large	C4	83% / 6.51 / 48.98	75% / 5.86 / 49.33	81% / 6.09 / 51.02	89% / 6.28 / 52.24	77% / 6.35 / 53.41	85% / 6.36 / 45.11
		C4	92% / 12.05 / 14.33	95% / 11.12 / 27.45	91% / 10.19 / 25.00	87% / 10.32 / 30.00	94% / 11.59 / 15.92	95% / 11.42 / 26.77

➤ Robustness: Success rate (SR) ↑ / Watermark Score ↑ / PPL ↓ **PVMark**

Watermarking Scheme	Attack	Benchmark	MiMC		Poseidon		Poseidon2
			two-to-one	three-to-one	two-to-one	three-to-one	two-to-one
KGW	Attack (a)	85.86% / 5.75 / 74.33	88.89% / 5.79 / 78.41	87.50% / 5.92 / 87.68	91.67% / 6.38 / 67.81	86.46% / 6.35 / 83.80	80.81% / 5.38 / 79.53
	Attack (b)	54.55% / 4.10 / 140.95	44.44% / 3.91 / 138.75	48.96% / 3.98 / 134.65	46.88% / 4.26 / 123.37	56.25% / 4.27 / 157.34	32.32% / 3.35 / 139.47
	Attack (c)	84.85% / 6.03 / 17.14	83.84% / 6.09 / 17.72	83.33% / 6.07 / 18.92	95.83% / 7.30 / 15.14	89.58% / 6.67 / 18.84	77.78% / 5.46 / 18.83

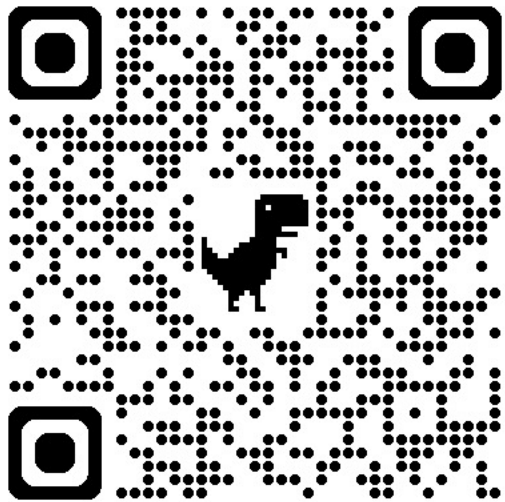
➤ Proof-generation cost (Setup time + Prove time)



2,000 tokens $\approx$ 30s

50,000 tokens $\approx$ 352s

Acceptable
Proof Overhead



Codes

Thanks !

Contact: duanhaohua@ecust.edu.cn

